

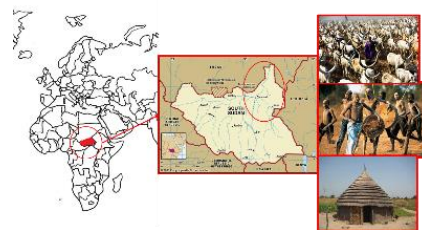
Your name:



The UK Linguistics Olympiad 2022 - Round 2

Problem 4. Dinka (25 marks)

Dinka is a Nilotic dialect cluster with about 1.3 million native speakers, mostly ethnic Dinka people in South Sudan. There are several main varieties, but this problem focuses on the Agar dialect.



When linguists first studied the language, they believed that the singular and plural forms of Dinka nouns were completely unpredictable. More recently, however, studying the way that verbs conjugate in Dinka allowed linguists to find patterns in the singular and plural forms of nouns. This has allowed many nouns to be grouped according to declension patterns, although many remain unexplained.

Below are 22 Dinka nouns, in both singular and plural forms, each of which follow a one of the common patterns. The translations have been provided only for interest: they have no bearing on the solution to the problem.

Note: **t**, **d**, **n**, **j** and **ŋ** are consonants; **ɛ** is the vowel in *bed*, and **ɔ** is the vowel in *not*. Dinka has three vowel lengths: **a** short, **aa** medium, and **aaa** long; as well as three tones, **á** high, **à** low, and **â** falling.

Singular	Plural	Translation	Singular	Plural	Translation	Singular	Plural	Translation
láj	làaj	<i>animal</i>	màac	mêec	<i>fire</i>	ḑél	ḑèel	<i>path</i>
gâaar	gêer	<i>ankle bell</i>	rúp	rwòop	<i>forest</i>	wáal	wál	<i>plant</i>
ḡaaar	ḡóor	<i>bean</i>	àdjéel	àdí	<i>gazelle</i>	ḡèeel	ḡéel	<i>python</i>
dít	djèet	<i>bird</i>	bôook	bóok	<i>hide</i>	dèenḡ	dêenḡ	<i>rain</i>
àḡwòòḡḡ	àḡóoḡḡ	<i>blacksmith</i>	ròok	ròok	<i>kidney</i>	àmàaal	àméel	<i>sheep</i>
ḡàanḡ	ḡêenḡ	<i>bow, gun</i>	ḡòòk	ḡòòk	<i>louse</i>	àtwòoor	àtúur	<i>slime</i>
ḡól	ḡaal	<i>cowdung fire</i>	àḡâaaḡ	àḡééḡ	<i>monitor lizard</i>			
twóòḡ	tónḡ	<i>egg</i>	àḡwòòk	àḡóòk	<i>monkey</i>			

Here are some forms of 4 regular Dinka verbs.

Root	1 st person	3 rd person	Translation
nòḡ	nàanḡ	nòòḡ	<i>to have</i>
kùc	kwòoc	kùuc	<i>to not know</i>
màat	màaat	mèèet	<i>to smoke</i>
lòòk	làaak	lòòk	<i>to wash</i>

Your name:



The UK Linguistics Olympiad 2022 - Round 2

4.1) Some singular or plural forms of Dinka nouns are given below. Only two of them follow one of the common patterns demonstrated above. On your answer sheet, mark which two they are.

Singular	Plural	Translation
àdèen		<i>beautiful one</i>
miiit		<i>firefly</i>
wèeet		<i>metal</i>
	tôon	<i>pot</i>
	jâak	<i>evil spirit</i>
	tûun	<i>horn</i>

4.2) Assuming that the following verbs conform to the previous pattern, fill in the correct form on your answer sheet:

Root	1 st person	3 rd person	Translation
a)	lwòccj	b)	<i>to be different</i>
c)	d)	cèem	<i>to eat</i>
e)	pèeεc	f)	<i>to loot</i>
wic	g)	wiic	<i>to need</i>
h)	i)	bòok	<i>to throw at</i>

4.3) Below are the singular or plural forms of 10 more Dinka nouns. By using the common patterns demonstrated before, predict the missing forms on your answer sheet. If there is more than one possible prediction, give them all.

Singular	Plural	Translation	Singular	Plural	Translation
ràaan	(j)	<i>person</i>	kók	o)	<i>hole in tree</i>
k)	wiil	<i>bristle</i>	p)	léek	<i>pestle</i>
rèeεc	l)	<i>fish</i>	ról	q)	<i>voice</i>
m)	kàal	<i>hole in ground</i>	jùit	r)	<i>well</i>
n)	rím	<i>blood</i>	ànâaar	s)	<i>buffalo</i>

4.4) Explain your solution.